



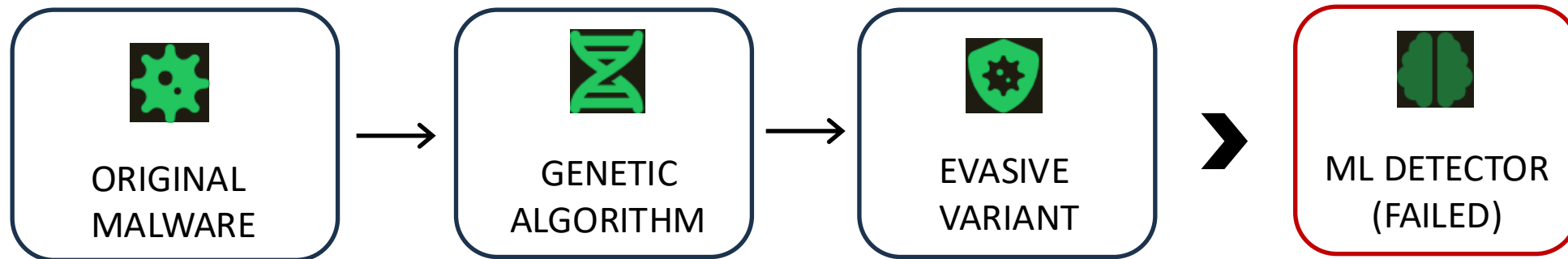
Towards Automated Generation of Adversarial Malware Samples with a Genetic Algorithm

József Sándor, Bence Kovács and Levente Buttyán

Laboratory of Cryptography and System Security (CrySyS Lab)
Budapest University of Technology and Economics (BME)

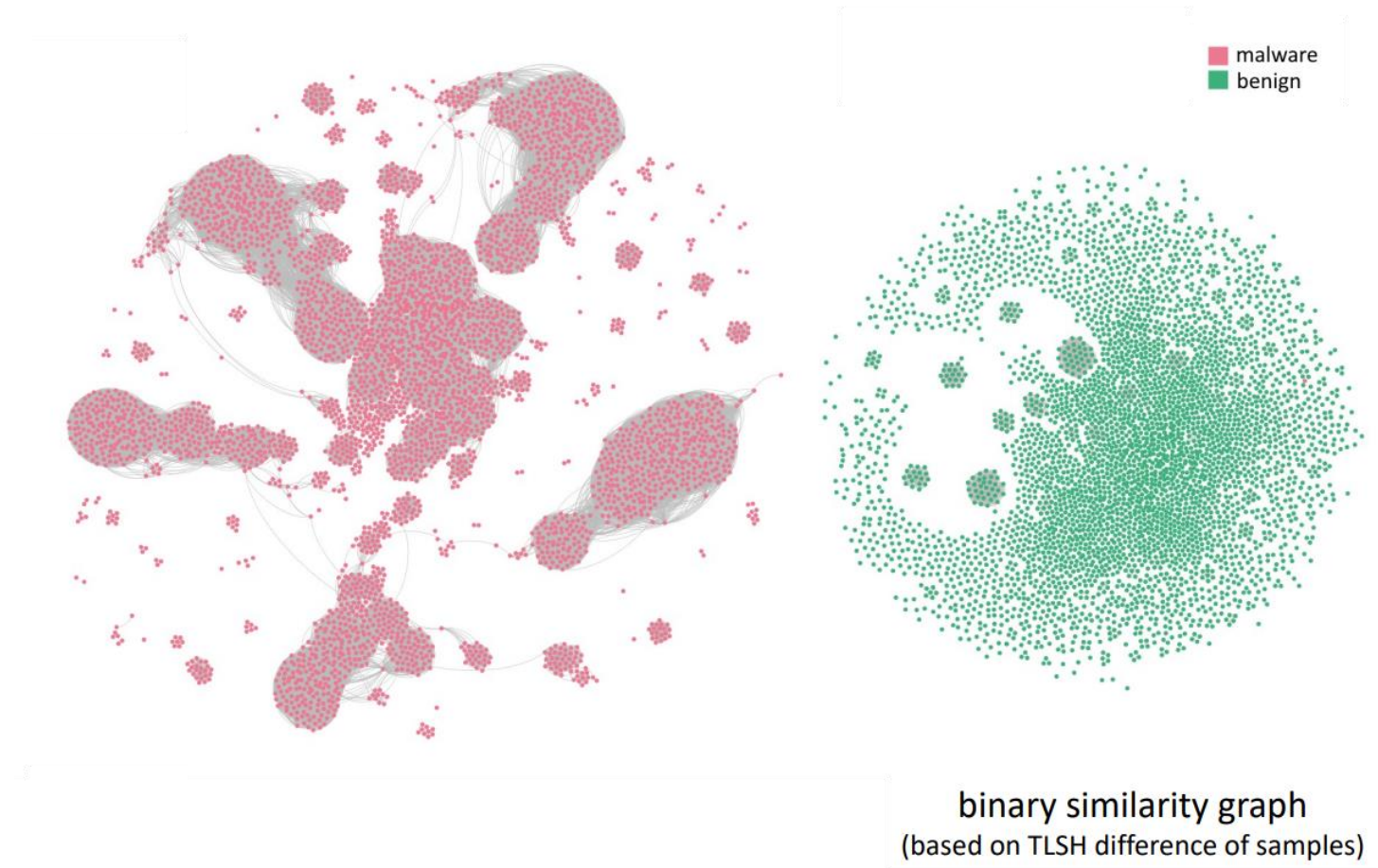
Overview

- Machine learning-based IoT malware detection
- Adversarial Malware Sample
 - specifically designed to cause the given model to make a mistake
 - same functionality as the original sample
- **GAME** framework
 - Genetic Algorithm for Malware Evason



SIMBioTA-ML

- ML-based IoT malware detection solution
 - Exploits binary similarity
- TLSH (Trend Micro Locality Sensitive Hash)
 - Similarity preserving hash function
 - Similarity score between two hashes
- TLSH hashes can be used as feature vectors for ML models
 - E.g., Random Forest Classifier
- Results
 - TPR > 95-97%
 - FPR < 1%
 - 140-160 kB model size



Dorottya Papp, Gergely Ács, Roland Nagy, and Levente Buttyán. SIMBioTA-ML: Lightweight, machine learning-based malware detection for embedded IoT devices. In IoTBDS, 2022.

Evasive Malware and Genetic Algorithm

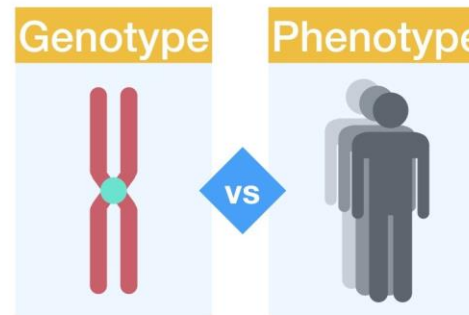
- Adversarial Malware Sample

- Avoiding the detection
- Same functionality as the original malware
- The modification to remain unnoticed
- Focusing on binary level modifications



- Population

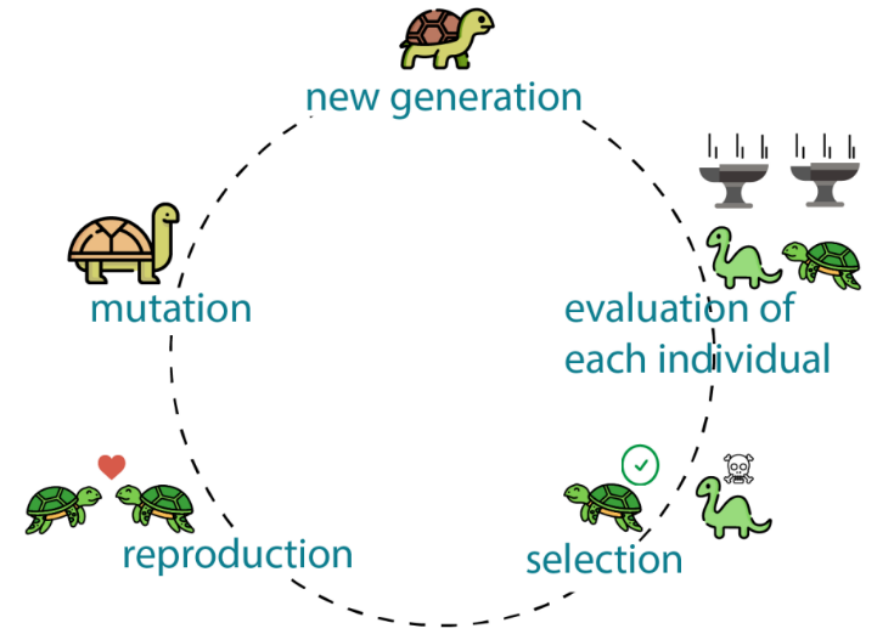
- Genotype -> modification strategy
 - » Contains genes
- Phenotype -> modified malware



- Fitness function -> attacker's preferences

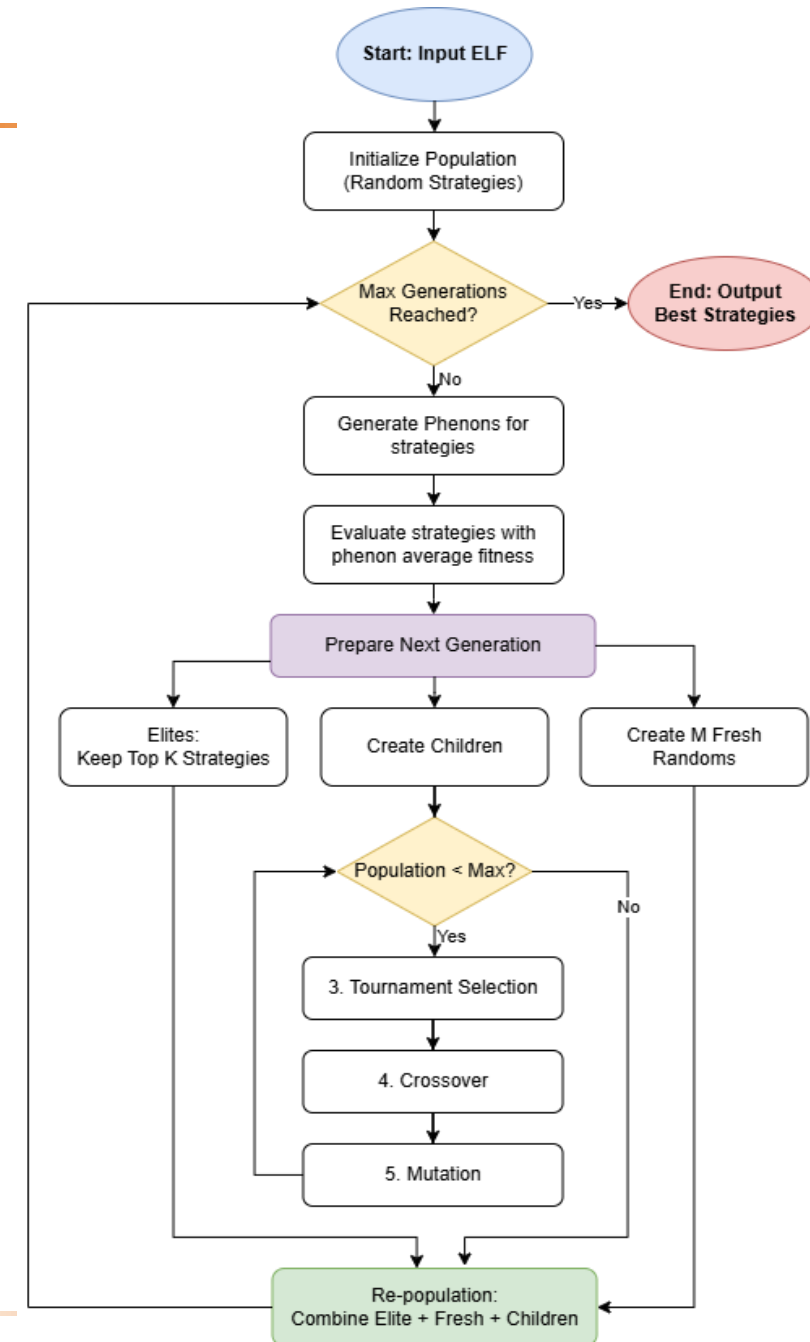
- Selection

- Crossover and mutation



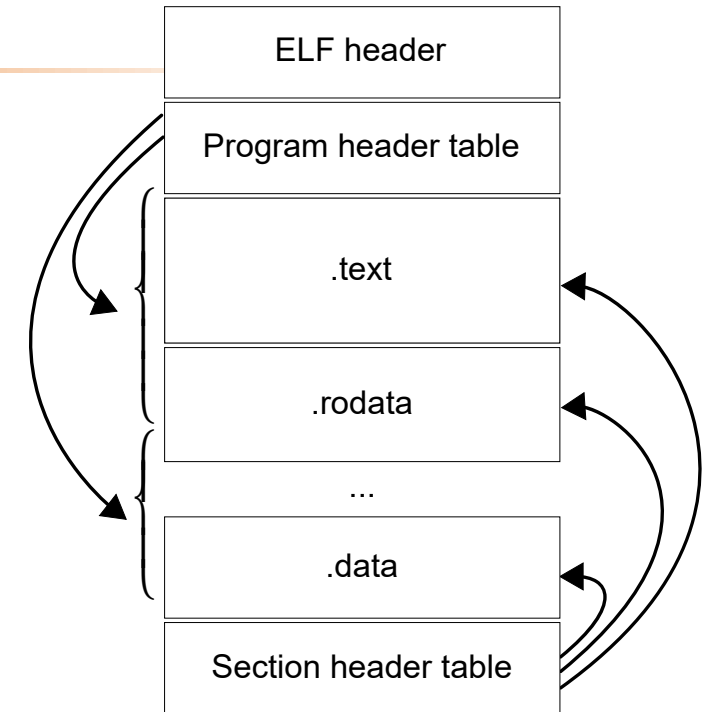
The Algorithm

1. Create N random strategies
2. Evaluate each strategy with the fitness function
3. New generation
 - Keep Top K Strategies
 - Extend with M new random strategy
 - Create children
 - » Tournament selection
 - » Crossover
 - » Mutation
4. Continue from step 2



ELF Modification Genes

- Appending to the end of the file (Appender)
 - Simple
 - Unlimited space
 - Easy to detect
- Padding overwriting (PaddingOverwriter)
 - What isn't loaded into memory is overwritten
 - 0% file size increase
 - Not trivial to detect
- New ELF segment injection (SegmentInjector)
 - Changes the structure of the file
 - Practically unlimited space
 - Hard to detect

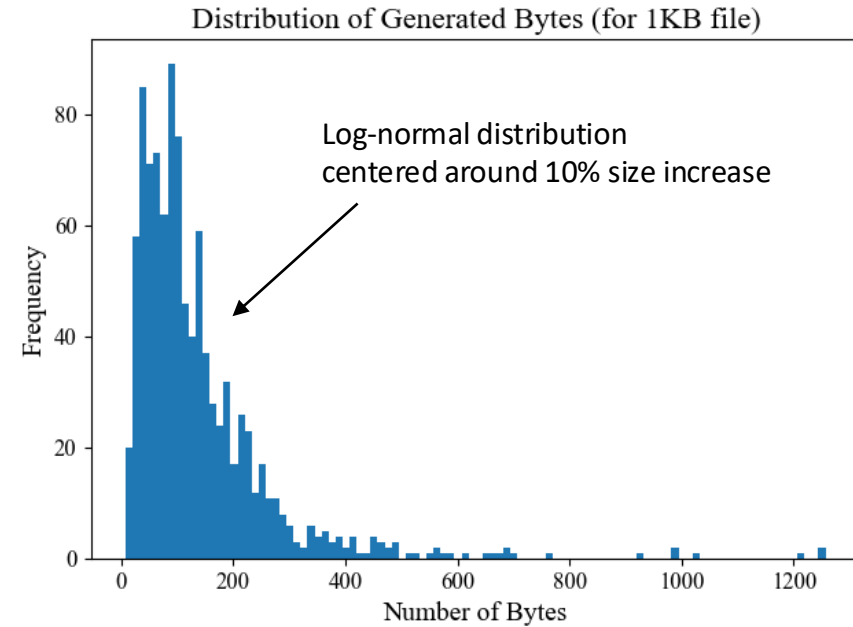
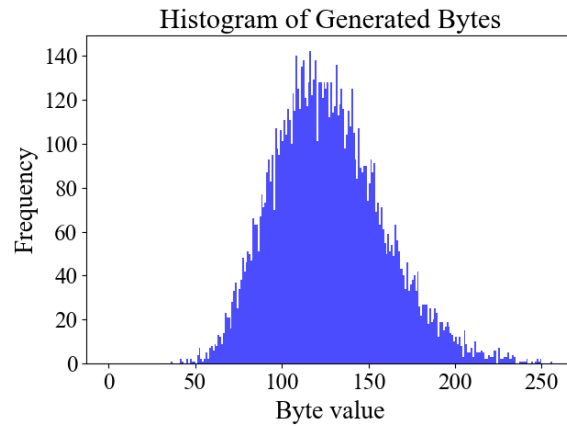


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Program Headers:
Type           Offset   VirtAddr   PhysAddr   FileSiz MemSiz  Flg Align
ARM_EXIDX      0x01bec0 0x0002bec0 0x0002bec0 0x00008 0x00008 R   0x4
PHDR           0x01becc 0x0002becc 0x0002becc 0x0402c 0x0402c R   0x4
INTERP         0x000154 0x00010154 0x00010154 0x00019 0x00019 R   0x1
    [Requesting program interpreter: /lib/ld-linux-armhf.so.3]
LOAD           0x000000 0x00010000 0x00010000 0x1fef8 0x1fef8 R E 0x10000
LOAD           0x01fef8 0x0003fef8 0x0003fef8 0x00410 0x01630 RW 0x10000
LOAD           0x022000 0x000a2000 0x000a2000 0x01000 0x01000 RWE 0x1000
DYNAMIC        0x01ff04 0x0003ff04 0x0003ff04 0x000d0 0x000d0 RW 0x4
NOTE           0x000170 0x00010170 0x00010170 0x00044 0x00044 R   0x4
GNU_STACK      0x000000 0x00000000 0x00000000 0x00000 0x00000 RW 0x10
GNU_RELRO      0x01fef8 0x0003fef8 0x0003fef8 0x00108 0x00108 R   0x1
```



Data Generation Genes

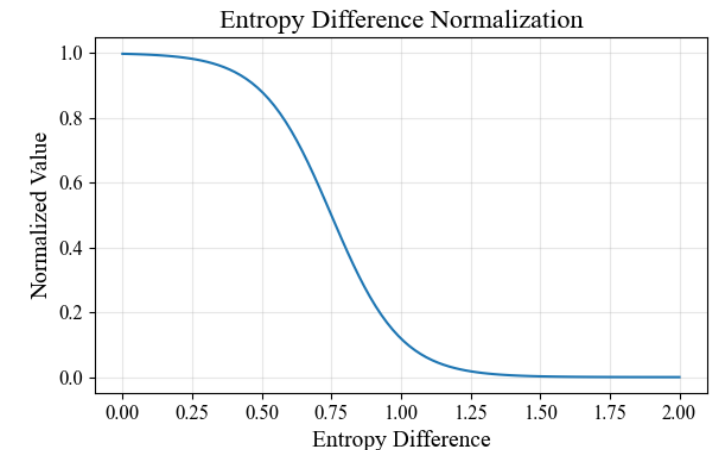
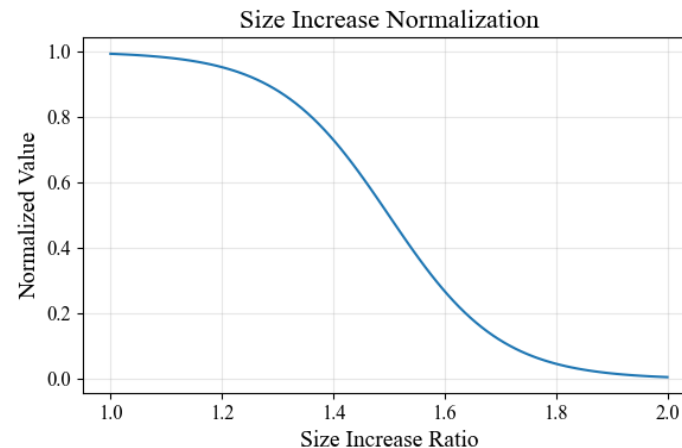
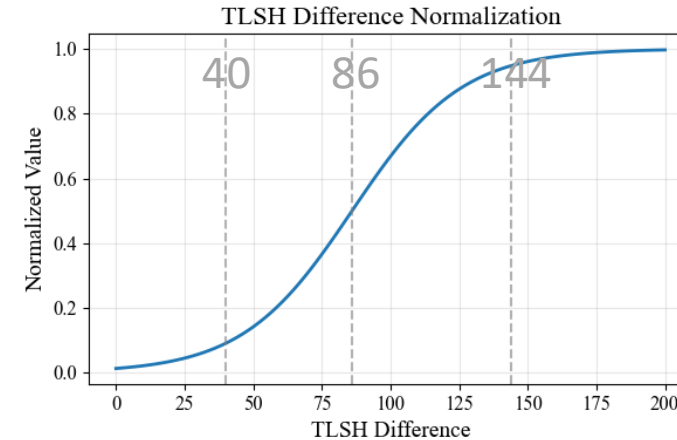
- Quantity of the generated data
- Generated data
 - Distribution-based: e.g., gamma



- File-based: sequential/random blocks
- Parameters are mutated during evolution

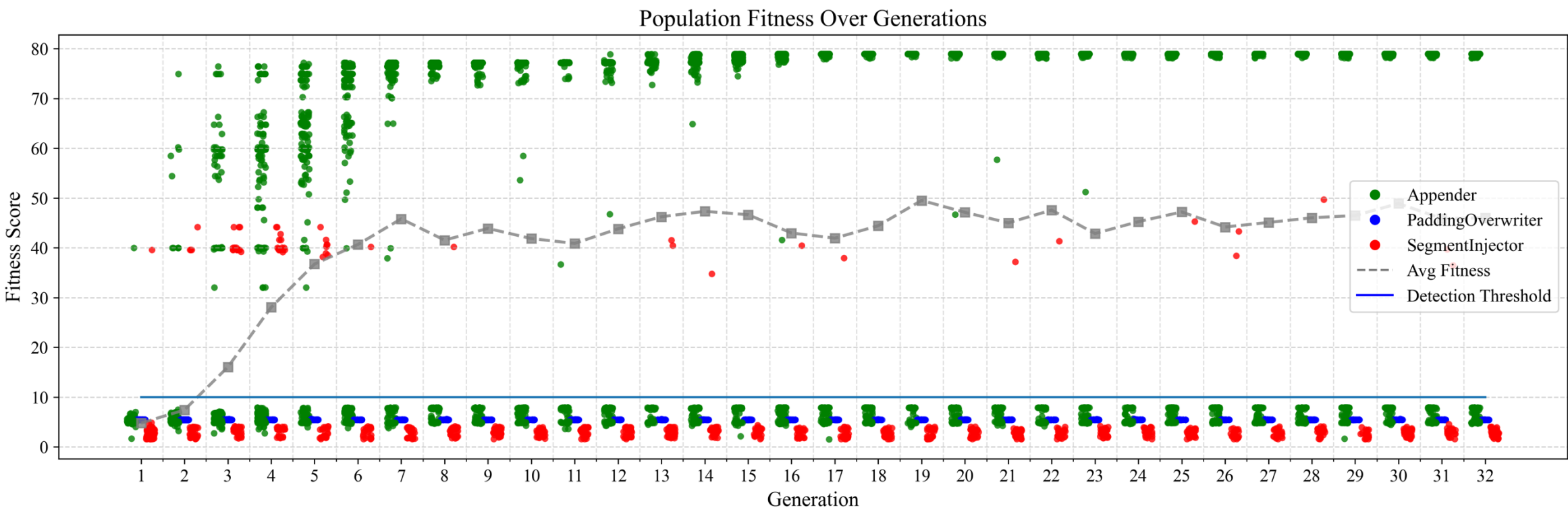
Fitness Function

- Encoding the preferences of the attacker
 - Score between 0-100
- Weighted sum of
 - TLSH difference (0.4)
 - File size increase (0.5)
 - Entropy difference (0.1)
- Anomaly weights
 - Appender (0.8)
 - PaddingOverwriter (0.9)
 - SegmentInjector (1.0)
- Detection penalty (**0.1**)



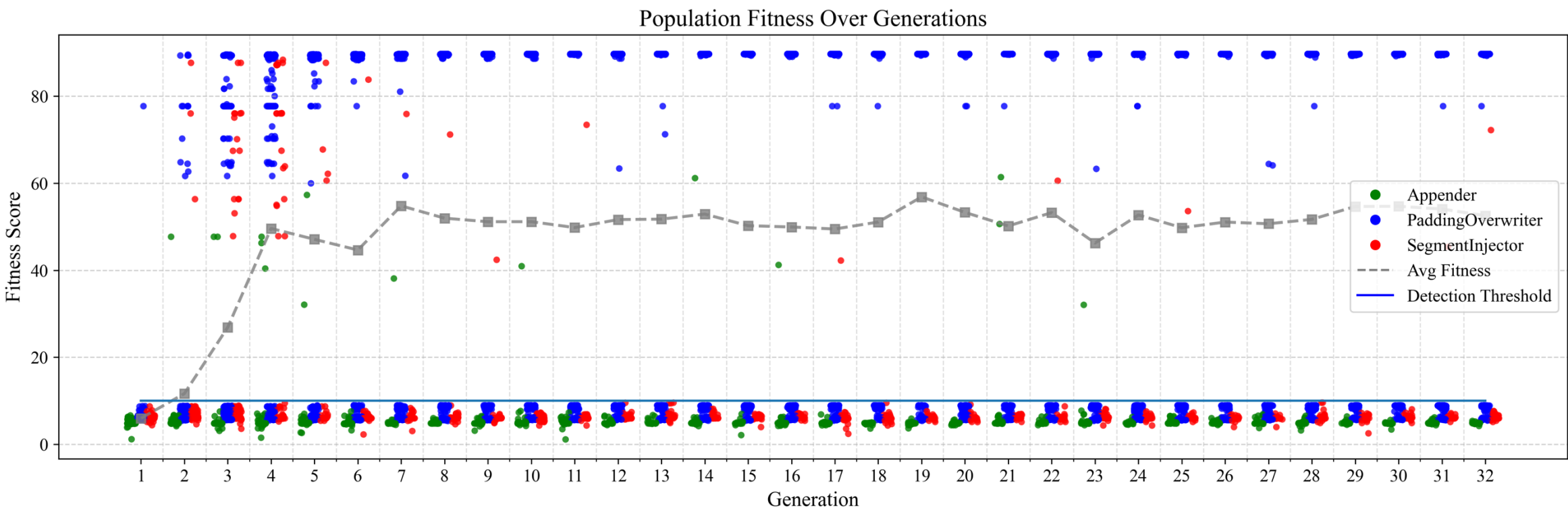
Results

For malware with SHA256: aee351932e...
+13% file size increase



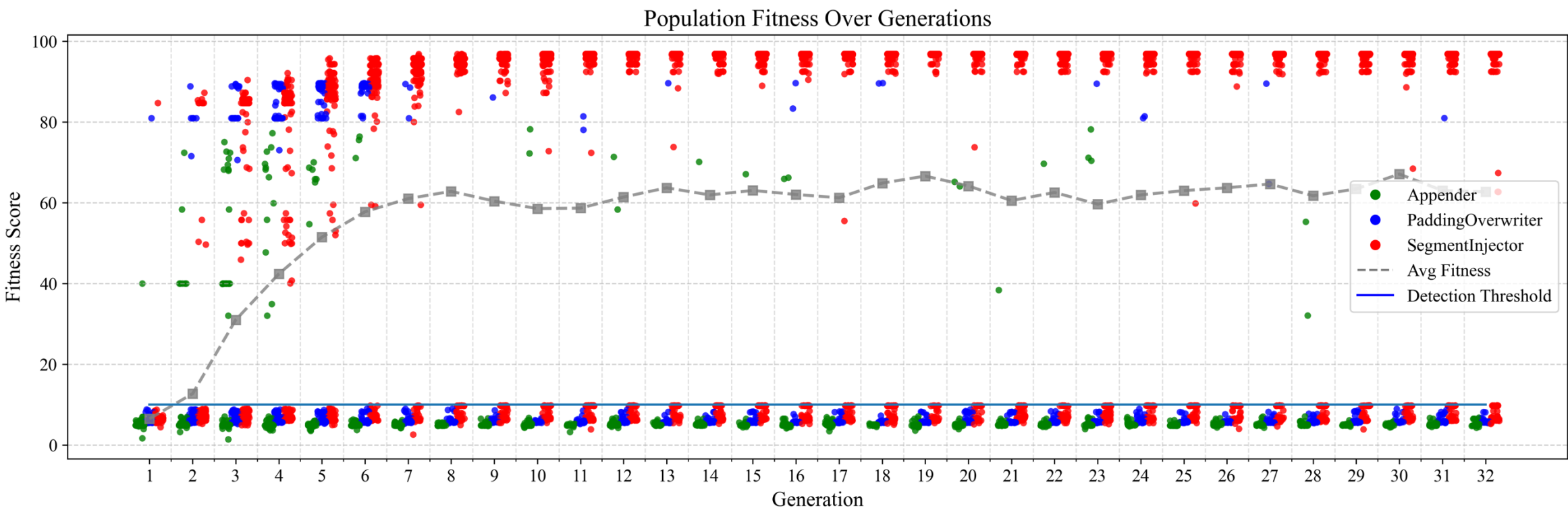
Results

For malware with SHA256: 68e1d2a46f...
+0% file size increase



Results

For malware with SHA256: 00eb525ad8...
+22% file size increase



Conclusion

- GAME framework is suitable to find adversarial modification strategies
- We can automatically create diverse and valid adversarial samples
- We can use them later for adversarial training